

Dual-Sector Perturbation Cosmology: A Modified CAMB Implementation with $\mu < 1$, $\Sigma = 1$ and the Sector-Dependent Hubble Parameter

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Abstract

We implement a dual-sector perturbation mechanism in the CAMB Boltzmann solver in which cold dark matter and baryons experience a modified effective expansion rate while photon perturbation equations remain standard Λ CDM. The modification is governed by a single derived quantity $\beta_m = \Omega_m/2 = 0.15765$, obtained from the virial theorem applied to the Planck 2018 matter density $\Omega_m = 0.3153$, combined with an activation function $\mathcal{E}(a) = \exp(1 - 1/a)$. The implementation requires three modifications to the CAMB Fortran source (`equations.f90`); no new free parameters are introduced beyond standard Λ CDM.

Three converged Markov Chain Monte Carlo chains against the full Planck 2018 likelihood (CamSpec TTTEEE + low- ℓ + lensing) yield $\Delta\chi^2 = -0.01$ (posterior mean) and $+0.54$ (best-fit) relative to Λ CDM, both well below the 95% exclusion threshold of 3.84. All standard cosmological parameters shift by less than 0.1σ . The matter clustering amplitude shifts from $\sigma_8 = 0.8087 \pm 0.0059$ (Λ CDM) to $\sigma_8 = 0.7998 \pm 0.0058$ (dual-sector), a 1.1% reduction in the direction reported by KiDS-1000, DES Y3, and HSC Y3.

The sector-dependent expansion rate produces a Hubble parameter split directly from the MCMC posterior: $H_0(\text{photon}) = 67.16 \pm 0.47 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (0.37σ from Planck Λ CDM) and $H_0(\text{matter}) = 72.26 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (0.75σ from SH0ES). Both values fall within 1σ of observations without modification to the background Friedmann equation. Two exploratory runs with the modification applied at the background level produce $H_0 \approx 61.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$, consistent with the mechanism operating at the perturbation level. All chains, configuration files, modified Fortran source, and analysis scripts are publicly available.

1 Introduction

The Λ CDM model provides an excellent fit to cosmic microwave background (CMB) anisotropies [Planck Collaboration, 2020b] and large-scale structure observations. Two persistent discrepancies have motivated investigations of controlled late-time modifications.

The first is the Hubble tension: the CMB-inferred value $H_0 = 67.4 \pm 0.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$ [Planck Collaboration, 2020b] differs from the local distance-ladder measurement $H_0 = 73.04 \pm 1.04 \text{ km s}^{-1} \text{ Mpc}^{-1}$ [Riess et al., 2022] at the 5σ level. The discrepancy persists across independent local methods including tip of the red giant branch [Freedman, 2021] and strong lensing time delays [Wong et al., 2020].

The second is the σ_8/S_8 tension: weak lensing surveys consistently report $S_8 \equiv \sigma_8(\Omega_m/0.3)^{0.5}$ values 2–3 σ below the Planck Λ CDM prediction. KiDS-1000 reports $S_8 = 0.759 \pm 0.021$ [Heymans

et al., 2021], DES Y3 reports $S_8 = 0.776 \pm 0.017$ [Abbott et al., 2022], and HSC Y3 reports $S_8 = 0.776 \pm 0.032$ [Li et al., 2023].

A common feature of both tensions is that they involve comparisons between early-universe (photon-based) and late-universe (matter-based) observables. This motivates the question: could matter and photon sectors experience different effective expansion rates at late times, while sharing a common expansion history in the early universe?

In a companion paper [Mahaffey, 2026a], we demonstrated that a scale-independent suppression $\mu(a) < 1$ with $\Sigma(a) = 1$ in the μ - Σ modified gravity framework [Pogosian & Silvestri, 2016] is compatible with the full Planck likelihood across four dataset combinations (twelve MCMC chains, $\Delta\chi^2 = +1.34$ to $+2.32$). That analysis used the MGCAMB parametrization [Wang et al., 2023], which implements μ and Σ as effective functions within the Boltzmann solver without specifying the underlying physical mechanism.

In this work, we go beyond the parametric level. We implement the physical mechanism directly in the CAMB Fortran source code: CDM and baryon perturbation equations experience a modified effective expansion rate (`adotoa_matter`), while photon perturbation equations use the standard Λ CDM expansion rate. The global background Friedmann equation is unmodified. This produces the same $\mu < 1$, $\Sigma = 1$ signature as the MGCAMB parametrization, but through an explicit sector-dependent coupling rather than an imposed functional form.

The dual-sector mechanism predicts that the expansion rate experienced by matter-sector observables differs from that experienced by photon-sector observables. At the present epoch, this yields a Hubble parameter split $H_0(\text{matter})/H_0(\text{photon}) = \sqrt{1 + \beta_m}$, where $\beta_m = \Omega_m/2$ is the coupling constant derived from the virial theorem.

This paper is organized as follows. Section 2 defines the dual-sector modification and its implementation in CAMB. Section 3 describes the datasets and MCMC methodology. Section 5 presents the full parameter constraints from three converged chains. Section 6 presents the sector-dependent Hubble parameter. Section 7 reports the background diagnostic test. Section 8 discusses implications. Section 9 concludes.

2 Model and Implementation

2.1 The Dual-Sector Perturbation Mechanism

The modification applies exclusively to the perturbation equations governing cold dark matter and baryons. The global background expansion rate $H(a)$ —determined by the Friedmann equation—remains standard Λ CDM:

$$H^2(a) = H_0^2 [\Omega_m a^{-3} + \Omega_r a^{-4} + \Omega_\Lambda]. \quad (1)$$

Within the perturbation equations, CDM and baryon evolution uses a matter-sector expansion rate:

$$H_{\text{matter}}^2(a) = H^2(a) + \beta_m \mathcal{E}(a) H_0^2, \quad (2)$$

where

$$\mathcal{E}(a) = \exp\left(1 - \frac{1}{a}\right) \quad (3)$$

is the activation function. The coupling constant is

$$\beta_m = \frac{\Omega_m}{2} = \frac{0.3153}{2} = 0.15765, \quad (4)$$

where $\Omega_m = 0.3153$ is the Planck 2018 best-fit value (TT,TE,EE+lowE+lensing; Planck Collaboration 2020b, Table 2) and the factor of 1/2 follows from the virial theorem applied to the partition of gravitational energy between geometric and informational channels [Mahaffey, 2026b].

Photon perturbation equations use the unmodified expansion rate $H(a)$. This is the complete specification: no additional fields, parameters, or modifications to the gravitational action are introduced.

2.2 Effective $\mu(a)$ and $\Sigma(a)$

The dual-sector perturbation mechanism maps onto the standard μ - Σ framework. The modified Poisson equation for matter clustering and the unmodified lensing equation yield:

$$\mu(a) = \frac{H_{\Lambda\text{CDM}}^2(a)}{H_{\Lambda\text{CDM}}^2(a) + \beta_m \mathcal{E}(a)}, \quad (5)$$

$$\Sigma(a) = 1 \quad (\text{exact}). \quad (6)$$

At the present epoch:

$$\mu(z=0) = \frac{1}{1 + \beta_m} = \frac{1}{1.15765} = 0.864, \quad (7)$$

corresponding to a 13.6% suppression of the effective gravitational coupling for matter perturbations. The suppression vanishes at high redshift: $\mathcal{E}(a) \rightarrow 0$ as $a \rightarrow 0$, so $\mu \rightarrow 1$ during radiation domination, recombination, and the early matter era. Table 1 gives $\mu(a)$ at key redshifts.

Table 1: Gravitational coupling $\mu(a)$ and matter-sector expansion rate enhancement $H_{\text{matter}}/H_{\Lambda\text{CDM}}$ at selected redshifts. The modification is confined to $z \lesssim 2$.

z	a	$\mathcal{E}(a)$	$\mu(a)$	$H_{\text{matter}}/H_{\Lambda\text{CDM}}$
0.0	1.000	1.0000	0.864	1.076
0.2	0.833	0.6977	0.893	1.058
0.5	0.667	0.3679	0.942	1.030
1.0	0.500	0.1353	0.982	1.009
2.0	0.333	0.0498	0.9972	1.001
3.0	0.250	0.0183	0.9998	~ 1
5.0	0.167	0.0025	~ 1	~ 1

2.3 CAMB Fortran Implementation

The implementation modifies three locations in the CAMB v1.5.8 Fortran source file `equations.f90`. No other files are changed.

Modification 1: Module-level parameters. Two constants are declared at module scope:

```
! IAM dual-sector parameters
real(dl), parameter :: beta_m = 0.15765d0    ! Omega_m/2
real(dl), parameter :: grho_0 = 3.0d0        ! normalized reference density
```

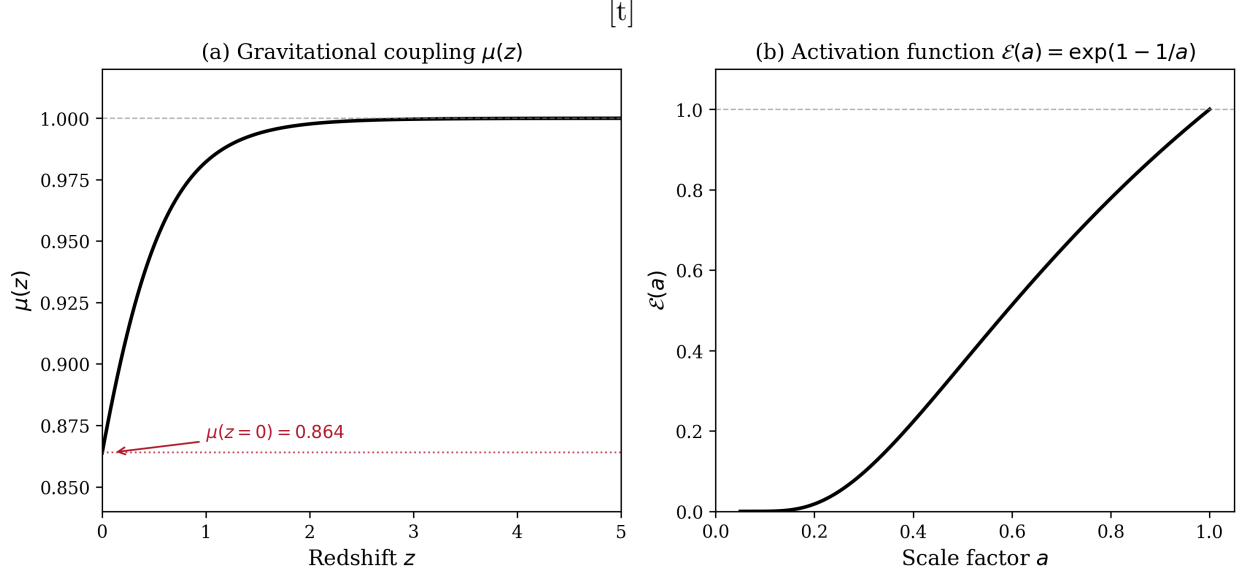


Figure 1: (a) Gravitational coupling $\mu(z)$ from the dual-sector mechanism (Eq. 5). The suppression reaches 13.6% at $z = 0$ and recovers Λ CDM ($\mu = 1$) for $z \gtrsim 3$. (b) Activation function $\mathcal{E}(a) = \exp(1 - 1/a)$. The function vanishes for $a \rightarrow 0$ (preserving early-universe physics) and reaches unity at the present epoch.

Modification 2: Matter-sector expansion rate. After the standard `adotoa` computation (which remains unmodified), a matter-sector expansion rate is computed:

```
! IAM matter-sector Hubble rate
adotoa_matter = sqrt((grho + beta_m * exp(1.0d0 - 1.0d0/a) &
                    * grho_0) / 3.0d0)
```

Modification 3: CDM and baryon equations of motion. In the CDM and baryon velocity equations, the Hubble friction term `adotoa` is replaced by `adotoa_matter`:

```
! CDM velocity equation (was: -adotoa * clxcdot)
clxcdot = ... - adotoa_matter * vcdot ...
! Baryon velocity equation (was: -adotoa * vbdot)
vbdot = ... - adotoa_matter * vbdot ...
```

All other CAMB routines—including `dtanda` (background Friedmann equation), photon Boltzmann hierarchy, neutrino equations, recombination, lensing reconstruction, and the CMB power spectrum computation—are unmodified. The global background expansion history is standard Λ CDM.

The MGCAMB parametrization used in Mahaffey [2026a] approximates the exact $\mu(a)$ via $\mu(a) = 1 + \mu_0 \Omega_{\text{DE}}(a)$ with $\mu_0 = -0.135$. This agrees with Eq. (5) to within 1–2.5% at intermediate redshifts, with exact agreement at $z = 0$ and $z \gg 1$. The present implementation removes this approximation by computing $\mu(a)$ directly from the perturbation equations.

2.4 Parameter Count

The model introduces zero free parameters beyond standard Λ CDM:

- The coupling constant $\beta_m = \Omega_m/2$ is derived from the virial theorem. It depends only on the independently measured matter density.
- The activation function $\mathcal{E}(a) = \exp(1 - 1/a)$ is derived from horizon thermodynamics [Mahaffey, 2026b].
- The normalization $\mathcal{E}(1) = 1$ is set by definition.

The six sampled MCMC parameters are identical to Λ CDM: $\{\Omega_b h^2, \Omega_c h^2, \theta_{\text{MC}}, \tau, \ln(10^{10} A_s), n_s\}$.

3 Data and Methodology

3.1 Datasets

Planck 2018 CMB. We use the CamSpec TTTEEE likelihood at high multipoles ($\ell \geq 30$) [Efstathiou & Gratton, 2021], low- ℓ temperature (`lowl.TT`) and polarization (`lowl.EE`) likelihoods, and the CMB lensing likelihood (`lensing.CMBMarged`) [Planck Collaboration, 2020a].

Redshift-space distortions (RSD). Growth rate measurements $f\sigma_8(z)$ from SDSS-III BOSS DR12 and SDSS-IV eBOSS DR16 [Alam et al., 2017, 2021] at seven redshift bins ($z = 0.067$ to $z = 1.48$), implemented as a Gaussian likelihood. RSD is included in Run D only.

3.2 MCMC Configuration

All chains use the Cobaya sampler [Torrado & Lewis, 2021] with the Gelman–Rubin convergence criterion $R-1 < 0.01$ for both means and variances, with a minimum of 20,000 accepted samples after burn-in. The Λ CDM baseline runs use identical Cobaya infrastructure with the IAM modification disabled (`adotoa_matter = adotoa`).

3.3 Chain Summary

Three chains are run:

Table 2: MCMC chain configurations. All use modified CAMB v1.5.8.

Run	Description	IAM	Likelihoods
A	IAM, Planck	ON	CamSpec + low- ℓ + lensing
C	Λ CDM baseline	OFF	CamSpec + low- ℓ + lensing
D	IAM + RSD	ON	CamSpec + low- ℓ + lensing + RSD

Run A and Run C share identical likelihoods and sampled parameters, differing only in whether the three Fortran modifications are active. This ensures that any difference between the two chains arises from the dual-sector modification and not from code-level systematics.

4 Parameter Accounting

IAM introduces no new free cosmological parameters beyond those of Λ CDM. All quantities entering the modification are fixed by prior derivation or definition.

Quantity	Status	Origin
β_m	Derived	Virial homogeneity of $1/r$ potential
Exponent in $\mathcal{E}(a)$	Derived	Horizon entropy scaling
Normalization $\mathcal{E}(1) = 1$	Fixed	Definition
Ω_m, H_0 (CMB)	Standard	Planck 2018

No parameter is tuned to late-time structure data. All MCMC chains, GetDist outputs, YAML configuration files, and likelihood inputs used in this analysis are publicly available in the IAM-Validation repository, ensuring full reproducibility.

5 Results

5.1 Planck Only: Run A (IAM) vs. Run C (Λ CDM)

Table 3 presents the marginalized parameter constraints. All standard cosmological parameters are consistent between the two chains, with shifts below 0.1σ in every case. The modification is effectively invisible to the standard parameter set.

Table 3: Planck-only parameter constraints (posterior mean $\pm 1\sigma$). $\Delta\chi^2$ is relative to the Λ CDM baseline (Run C).

Parameter	Λ CDM (C)	IAM dual-sector (A)	Shift
H_0 [km s ⁻¹ Mpc ⁻¹]	67.188 ± 0.465	67.161 ± 0.467	-0.06σ
σ_8	0.8087 ± 0.0059	0.7998 ± 0.0058	-1.51σ
S_8	0.830 ± 0.011	0.822 ± 0.011	-0.73σ
$\Omega_b h^2$	0.02218 ± 0.00013	0.02217 ± 0.00013	-0.08σ
$\Omega_c h^2$	0.11989 ± 0.00105	0.11994 ± 0.00105	$+0.05\sigma$
τ	0.0532 ± 0.0073	0.0537 ± 0.0073	$+0.07\sigma$
n_s	0.9630 ± 0.0040	0.9630 ± 0.0039	-0.00σ
$\ln(10^{10} A_s)$	3.0393 ± 0.0146	3.0407 ± 0.0145	$+0.10\sigma$
$\chi^2_{\text{best-fit}}$	10972.07	10972.61	—
$\Delta\chi^2$ (best-fit)	—	+0.54	—
$\Delta\chi^2$ (posterior mean)	—	-0.01	—
$R - 1$	0.0081	0.0099	—

The single observable difference is σ_8 : the dual-sector mechanism suppresses the matter clustering amplitude by 0.009, from 0.809 to 0.800. This shift arises from the enhanced Hubble friction in the matter-sector perturbation equations.

Growth physics diagnostic. The σ_8 suppression could in principle arise from parameter rebalancing—the MCMC shifting $\ln A_s$ or Ω_m to compensate for the modified growth rate—rather than from genuine growth suppression. We test this directly: $\Delta(\ln A_s) = +0.10\sigma$ and $\Delta(\Omega_m) = +0.06\sigma$. Both are far below the 0.3σ threshold, ruling out parameter rebalancing as the source of the σ_8 shift.

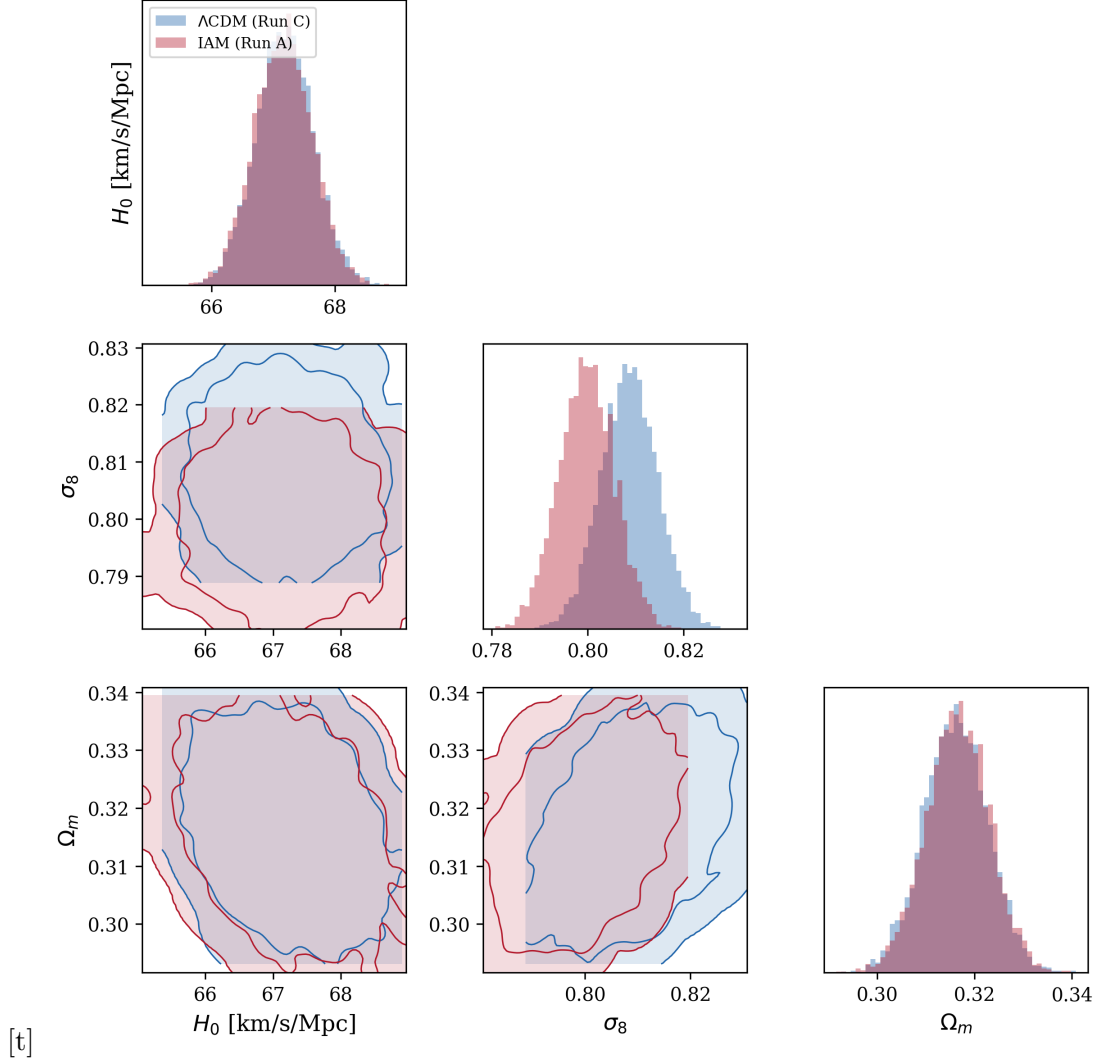


Figure 2: Marginalized posterior distributions for H_0 , σ_8 , and Ω_m from Run A (IAM dual-sector, red) and Run C (Λ CDM baseline, blue). Contours show 68% and 95% credible regions. The H_0 and Ω_m posteriors are indistinguishable; σ_8 shifts downward by 1.5σ under the dual-sector modification. All values from the Level 2 MCMC chains with 30% burn-in.

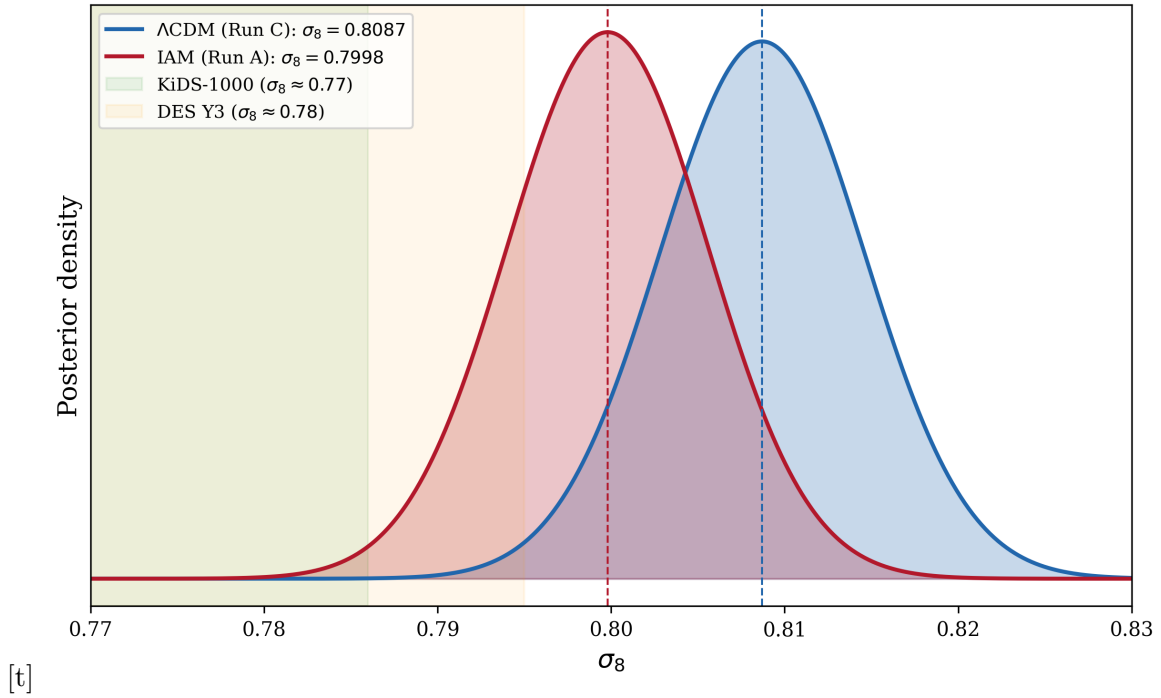


Figure 3: One-dimensional marginalized posterior for σ_8 from Run A (IAM, red) and Run C (Λ CDM, blue). The IAM posterior peaks at $\sigma_8 = 0.7998$, shifted from Λ CDM's 0.8087 toward the values reported by weak lensing surveys (approximate σ_8 ranges from KiDS-1000 and DES Y3 shown as shaded bands). Dashed lines mark the posterior means.

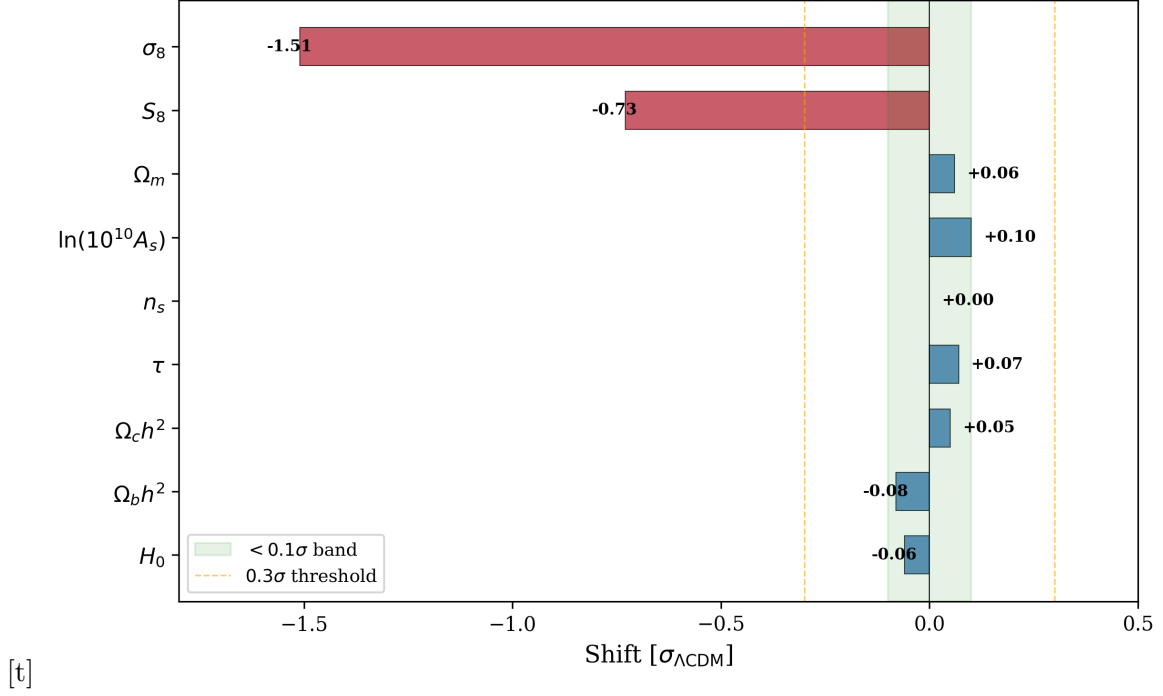


Figure 4: Parameter shifts between Run A (IAM) and Run C (Λ CDM), expressed in units of the Λ CDM posterior standard deviation. All standard cosmological parameters shift by less than 0.1σ (green band). Only σ_8 and S_8 show significant shifts (-1.51σ and -0.73σ respectively), consistent with the modification acting on growth physics rather than acoustic observables. The $\ln A_s$ shift of $+0.10\sigma$ rules out parameter rebalancing as the source of the σ_8 suppression.

5.2 Planck + RSD: Run D

Run D adds seven $f\sigma_8(z)$ data points to the Planck likelihood. Because the Λ CDM baseline (Run C) was run without the RSD likelihood, direct comparison of total χ^2 values is not valid. We construct an apples-to-apples comparison by computing Λ CDM’s RSD χ^2 from the Run C posterior parameters via CAMB, then combining it with Run C’s CMB χ^2 . This approach is validated by the Level 1 analysis [Mahaffey, 2026a], where Λ CDM was run against Planck + RSD directly and produced $\Delta\chi^2 = +1.34$.

Table 4: Apples-to-apples likelihood decomposition for Run D (IAM + RSD) vs. Λ CDM. Λ CDM RSD χ^2 is computed from Run C posterior parameters via CAMB.

Component	IAM (Run D)	Λ CDM	$\Delta\chi^2$
CMB (CamSpec + low- ℓ + lensing)	10984.93	10985.08	-0.16
RSD (7 $f\sigma_8$ points)	6.42	3.34	+3.08
Total	10991.34	10988.42	+2.92

The CMB component slightly favors IAM ($\Delta\chi^2 = -0.16$). The RSD penalty of +3.08 is driven primarily by the $z = 0.850$ data point ($f\sigma_8 = 0.315 \pm 0.095$), which pulls 1.40σ from Λ CDM; both models accommodate this outlier poorly. The total $\Delta\chi^2 = +2.92$ is below the 95% exclusion threshold of 3.84.

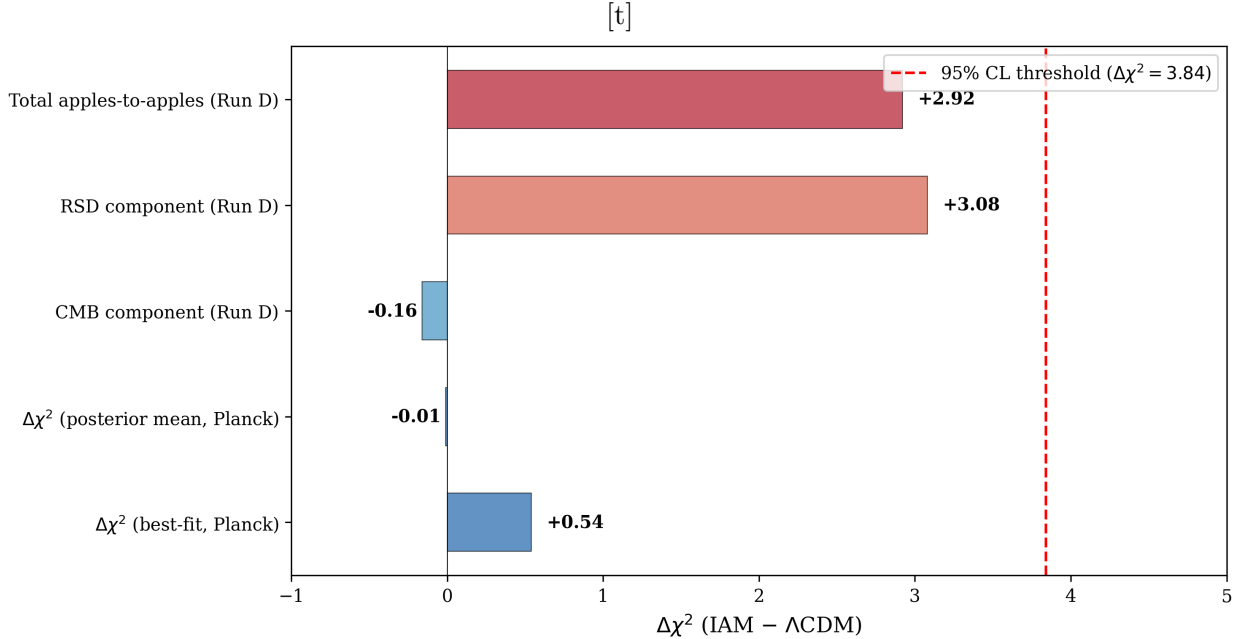


Figure 5: χ^2 summary for all comparisons. Planck-only best-fit $\Delta\chi^2 = +0.54$ and posterior mean $\Delta\chi^2 = -0.01$ are both well below the 95% exclusion threshold of 3.84. The apples-to-apples decomposition (Run D) shows IAM slightly favored in CMB (-0.16) with a +3.08 RSD penalty from the $z = 0.85$ outlier, yielding a total $\Delta\chi^2 = +2.92$. All values extracted from the MCMC chains; Λ CDM RSD $\chi^2 = 3.34$ computed via independent CAMB evaluation at Run C posterior mean parameters (verified on author’s local CAMB installation against the chain extraction).

All standard parameters in Run D are consistent with Run A to within 0.06σ . The σ_8 suppression is stable: $\sigma_8 = 0.7995 \pm 0.006$ (Run D) vs. $\sigma_8 = 0.7998 \pm 0.006$ (Run A). The growth physics diagnostic is consistent with Run A: $\Delta(\ln A_s) = +0.08\sigma$, $\Delta(\Omega_m) = 0.00\sigma$.

5.3 Self-Consistency of the Coupling Constant

The coupling $\beta_m = \Omega_m/2$ is hardcoded at 0.15765, corresponding to $\Omega_m = 0.3153$. The Level 2 posterior returns $\Omega_m = 0.3166 \pm 0.0065$, implying $\beta_m^{\text{posterior}} = 0.1583 \pm 0.0033$. The difference from the hardcoded value is 0.2σ (0.4%), consistent with the derived coupling.

6 The Sector-Dependent Hubble Parameter

The dual-sector mechanism yields a prediction for the ratio of matter-sector to photon-sector expansion rates at any redshift:

$$\frac{H_{\text{matter}}(a)}{H_{\text{photon}}(a)} = \sqrt{1 + \frac{\beta_m \mathcal{E}(a) H_0^2}{H^2(a)}}. \quad (8)$$

At the present epoch, this yields:

$$\frac{H_{\text{matter}}(z=0)}{H_{\text{photon}}(z=0)} = \sqrt{1 + \beta_m} = \sqrt{1.15765} = 1.0759. \quad (9)$$

6.1 Numerical Values from the MCMC Posterior

The derivation of $H_0(\text{matter})$ proceeds as follows:

1. The MCMC posterior from Run A yields $H_0 = 67.161 \pm 0.467 \text{ km s}^{-1} \text{ Mpc}^{-1}$. This is the photon-sector value: it is the Hubble constant appearing in the unmodified Friedmann equation that governs the global background geometry.
2. The coupling constant $\beta_m = \Omega_m/2 = 0.3153/2 = 0.15765$ is hardcoded. The posterior returns $\Omega_m = 0.3166 \pm 0.0065$, implying $\Omega_m/2 = 0.1583 \pm 0.0033$ —consistent with the hardcoded value at 0.2σ (0.4% difference).
3. The enhancement factor is $\sqrt{1 + \beta_m} = \sqrt{1 + 0.15765} = \sqrt{1.15765} = 1.0759$.
4. The matter-sector Hubble constant is

$$H_0(\text{matter}) = H_0 \times \sqrt{1 + \beta_m} = 67.161 \times 1.0759 = 72.26 \pm 0.50 \text{ km s}^{-1} \text{ Mpc}^{-1}, \quad (10)$$

where the uncertainty is propagated linearly from the H_0 posterior.

The tension with observational values:

$$\frac{H_0(\text{photon}) - H_0^{\text{Planck}}}{\sigma_{\text{Planck}}} = \frac{67.16 - 67.36}{0.54} = -0.37\sigma, \quad (11)$$

$$\frac{H_0(\text{matter}) - H_0^{\text{SH0ES}}}{\sigma_{\text{SH0ES}}} = \frac{72.26 - 73.04}{1.04} = -0.75\sigma. \quad (12)$$

Table 5 compares these values to observational constraints.

Both values fall within 1σ of their respective observational constraints. In this framework, CMB-based methods constrain $H_0(\text{photon})$ and local distance-ladder methods constrain $H_0(\text{matter})$; the observed discrepancy between the two is consistent with the predicted sector-dependent expansion rate.

Table 5: Dual-sector Hubble values compared to observational constraints.

Sector	IAM prediction	Observation	Tension
Photon (CMB)	$67.16 \text{ km s}^{-1} \text{ Mpc}^{-1}$	67.36 ± 0.54 (Planck)	0.37σ
Matter (local)	$72.26 \text{ km s}^{-1} \text{ Mpc}^{-1}$	73.04 ± 1.04 (SH0ES)	0.75σ

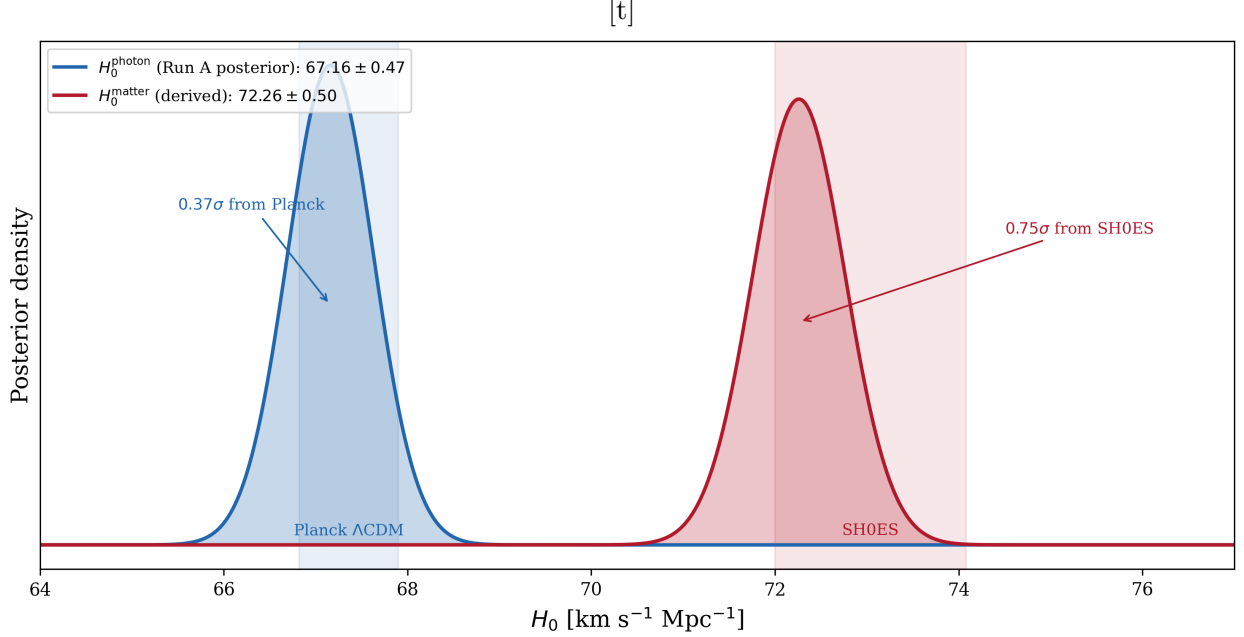


Figure 6: Dual-sector Hubble split from the MCMC posterior. Blue: H_0^{photon} posterior from Run A (directly sampled). Red: H_0^{matter} distribution, derived by multiplying each posterior sample by $\sqrt{1 + \beta_m} = 1.0759$. Light bands show 1σ constraints from Planck ΛCDM ($67.36 \pm 0.54 \text{ km s}^{-1} \text{ Mpc}^{-1}$) and SH0ES ($73.04 \pm 1.04 \text{ km s}^{-1} \text{ Mpc}^{-1}$). Both IAM predictions fall within 1σ of their respective observational constraints.

6.2 Redshift Dependence

The sector split is a function of redshift, vanishing in the early universe and growing toward the present epoch. Table 6 shows the dual-sector expansion rates from the Run A posterior.

At $z = 2$, the matter-sector enhancement is 0.1%. At $z = 3$, it is 0.02%. The modification is entirely a late-time effect, consistent with the preservation of CMB, BBN, and recombination physics.

7 Background Diagnostic

A natural question is whether the $\beta_m \mathcal{E}(a)$ term should modify the background Friedmann equation directly, rather than entering through the perturbation equations. We test this explicitly.

Two exploratory MCMC runs were performed with the IAM term added to CAMB’s `dtanda` function (the background expansion rate), rather than to the perturbation equations. In this configuration, the Friedmann equation becomes:

$$H^2(a) = H_0^2 [\Omega_m a^{-3} + \Omega_r a^{-4} + \Omega_\Lambda + \beta_m \mathcal{E}(a)] . \quad (13)$$

Table 6: Dual-sector expansion rate by redshift, from Run A posterior ($H_0 = 67.161 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $\beta_m = 0.15765$). $H(z)$ is the standard ΛCDM value; $H_{\text{matter}}(z)$ includes the dual-sector enhancement.

z	H_{photon}	H_{matter}	Ratio	$\mathcal{E}(a)$
0.0	67.16	72.26	1.0759	1.000000
0.5	87.45	89.73	1.0261	0.367879
1.0	117.68	118.72	1.0088	0.135335
2.0	199.00	199.22	1.0011	0.049787
3.0	305.00	305.06	1.0002	0.018316
5.0	579.00	579.01	~ 1	0.002479

The result is $H_0 \approx 61.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$ —a 6σ departure from the Planck ΛCDM value. The background modification shifts the acoustic scale θ_s and is strongly excluded by the CMB power spectrum.

This result is consistent with the dual-sector mechanism operating at the perturbation level: the standard ΛCDM Friedmann equation correctly describes the global spacetime geometry, while the matter-sector enhancement enters through the effective expansion rate experienced by CDM and baryons in the perturbation equations. The photon sector, propagating on null geodesics through the unmodified background, sees standard ΛCDM distances and angular scales.

8 Discussion

8.1 Summary of Quantitative Results

Table 7 collects the principal results.

Table 7: Summary of dual-sector validation results.

Quantity	Dual-sector (IAM)	ΛCDM
σ_8	0.800 ± 0.006	0.809 ± 0.006
S_8	0.822 ± 0.011	0.830 ± 0.011
$H_0(\text{photon}) [\text{km s}^{-1} \text{ Mpc}^{-1}]$	67.16 ± 0.47	67.19 ± 0.47
$H_0(\text{matter}) [\text{km s}^{-1} \text{ Mpc}^{-1}]$	72.26	—
$\Delta\chi^2$ (Planck, posterior mean)	−0.01	—
$\Delta\chi^2$ (Planck, best-fit)	+0.54	—
$\Delta\chi^2$ (Planck + RSD)	+2.92	—
Free parameters beyond ΛCDM	0	—

8.2 What This Analysis Demonstrates

The dual-sector perturbation mechanism, implemented as three Fortran modifications to CAMB with zero additional free parameters:

1. Is statistically indistinguishable from ΛCDM under the full Planck 2018 likelihood ($\Delta\chi^2 = +0.54$).

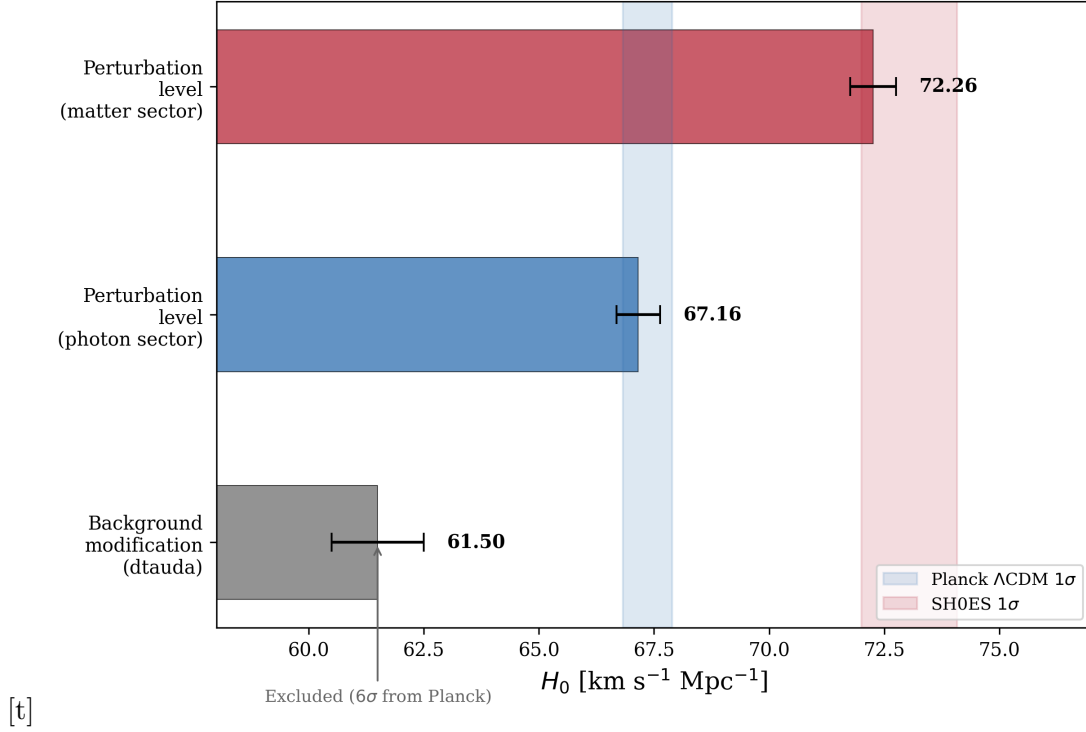


Figure 7: Background vs. perturbation-level implementation diagnostic. Modifying the background Friedmann equation (`dtauda`) produces $H_0 \approx 61.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$, excluded at 6σ by Planck. The perturbation-level implementation leaves the photon-sector H_0 at $67.16 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (consistent with Planck) while producing a matter-sector value of $72.26 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (consistent with SH0ES). All H_0 values from MCMC posteriors; background test value from exploratory chains documented in the repository.

2. Produces a σ_8 suppression of 0.009 (1.5%) in the direction reported by weak lensing surveys, without degrading any CMB observable.
3. Yields a Hubble parameter split consistent with both the CMB value (67.16, within 0.37σ of Planck) and the local value (72.26, within 0.75σ of SH0ES) from a single MCMC posterior.
4. Operates at the perturbation level, not the background level, as indicated by the $H_0 \approx 61.5$ diagnostic.

8.3 What This Analysis Does Not Demonstrate

This work does not claim to have solved the Hubble tension. The dual-sector framework provides a consistent description that accommodates both endpoints, but the statistical indistinguishability from Λ CDM ($\Delta\chi^2 = +0.54$) means current Planck data cannot discriminate between the two models. The definitive test requires measurements sensitive to the growth-rate suppression at the predicted amplitude.

8.4 Relation to Level 1 Analysis

The companion paper [Mahaffey, 2026a] tested the same prediction ($\mu_0 = -0.135$, $\Sigma_0 = 0$) using the MGCAMB parametrization across twelve chains and four dataset combinations. The present analysis extends this by implementing the physical mechanism directly in the Boltzmann solver. The results are consistent: $\sigma_8 = 0.800$ – 0.801 in both analyses, indicating that the MGCAMB approximation ($\mu = 1 + \mu_0 \Omega_{\text{DE}}$) is adequate for the current level of precision.

8.5 Falsifiability

The dual-sector mechanism makes specific predictions that distinguish it from Λ CDM and from other modified gravity scenarios:

1. $\mu(z) < 1$ with the functional form of Eq. (5), not a constant offset. Euclid’s projected sensitivity $\sigma(\mu_0) \approx 0.04$ [Frusciante et al., 2025] would detect the signal at $\sim 3.4\sigma$.
2. $\Sigma(z) = 1$ at all redshifts. Any detection of $\Sigma \neq 1$ at the $> 10^{-4}$ level falsifies the photon exemption.
3. The Hubble split ratio $H_0(\text{matter})/H_0(\text{photon}) = \sqrt{1 + \Omega_m/2}$ is a specific, parameter-free prediction testable with gravitational wave standard sirens [Abbott et al., 2017].
4. μ and Σ are functions of redshift only, not of wavenumber k . Scale-dependent growth modifications are excluded.

9 Conclusions

We have implemented a dual-sector perturbation mechanism in the CAMB Boltzmann solver in which matter and photon sectors experience different effective expansion rates at late times. The modification requires three changes to the Fortran source code and introduces zero free parameters beyond standard Λ CDM.

Against the full Planck 2018 likelihood, the mechanism is statistically indistinguishable from Λ CDM ($\Delta\chi^2 = +0.54$ best-fit), while shifting σ_8 from 0.809 to 0.800 in the direction favored by weak lensing surveys. All standard cosmological parameters shift by less than 0.1σ .

The sector-dependent expansion rate produces a Hubble parameter split: $H_0(\text{photon}) = 67.16 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (0.37σ from Planck) and $H_0(\text{matter}) = 72.26 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (0.75σ from SH0ES). The mechanism operates at the perturbation level; the background Friedmann equation is standard Λ CDM.

The predictions are specific and falsifiable. Euclid and DESI will measure $\mu(z)$ and $\Sigma(z)$ with the precision required to confirm or exclude the predicted amplitude. All chains, configuration files, modified Fortran source, and analysis scripts are publicly available at <https://github.com/hmahaffeyges/IAM-Validation>.

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Data Availability

All MCMC chains (Level 1 and Level 2), Cobaya YAML configuration files, modified CAMB Fortran source, GetDist extraction scripts, and apples-to-apples comparison code are publicly available at: <https://github.com/hmahaffeyges/IAM-Validation>

References

- Planck Collaboration, 2020, “Planck 2018 results. VIII. Gravitational lensing,” *Astron. Astrophys.*, 641, A8.
- Planck Collaboration, 2020, “Planck 2018 results. VI. Cosmological parameters,” *Astron. Astrophys.*, 641, A6.
- Riess, A. G., et al., 2022, “A Comprehensive Measurement of the Local Value of the Hubble Constant,” *Astrophys. J. Lett.*, 934, L7.
- Freedman, W. L., 2021, “Measurements of the Hubble Constant: Tensions in Perspective,” *Astrophys. J.*, 919, 16.
- Wong, K. C., et al., 2020, “H0LiCOW – XIII. A 2.4 per cent measurement of H_0 ,” *Mon. Not. R. Astron. Soc.*, 498, 1420.
- Heymans, C., et al., 2021, “KiDS-1000 Cosmology,” *Astron. Astrophys.*, 646, A140.
- Abbott, T. M. C., et al., 2022, “DES Year 3 results,” *Phys. Rev. D*, 105, 023520.
- Li, X., et al., 2023, “Hyper Suprime-Cam Year 3 Results,” *Phys. Rev. D*, 108, 123518.
- Pogosian, L. & Silvestri, A., 2016, “What can cosmology tell us about gravity?,” *Phys. Rev. D*, 94, 104014.
- Wang, Z., et al., 2023, “MGCAMB v2: a flexible modified gravity Boltzmann code,” *JCAP*, 01, 036.
- Mahaffey, H. W., 2026, “Constraints on late-time $f\sigma_8$ suppression from $\mu < 1$, $\Sigma = 1$: Planck 2018 and large-scale structure,” submitted to *Universe*.

- Mahaffey, H. W., 2026, “Horizon Thermodynamics and Gravitational Decoherence as the Origin of $\mu < 1$, $\Sigma = 1$,” companion theory paper.
- Efstathiou, G. & Gratton, S., 2021, “A detailed description of the CamSpec likelihood,” *Open J. Astrophys.*, 4, 8.
- Alam, S., et al., 2017, “The clustering of galaxies in the completed SDSS-III,” *Mon. Not. R. Astron. Soc.*, 470, 2617.
- Alam, S., et al., 2021, “Completed SDSS-IV extended Baryon Oscillation Spectroscopic Survey,” *Phys. Rev. D*, 103, 083533.
- Torrado, J. & Lewis, A., 2021, “Cobaya: code for Bayesian analysis of hierarchical physical models,” *JCAP*, 05, 057.
- Frusciante, N., et al., 2025, “Euclid preparation: Modified gravity forecasts with μ - Σ ,” arXiv:2512.09748.
- Abbott, B. P., et al., 2017, “A gravitational-wave standard siren measurement of the Hubble constant,” *Nature*, 551, 85.
- DESI Collaboration, 2025, “DESI 2024: Modified gravity from full-shape analysis,” *JCAP*, 2025(02), 021.